

Machine Learning Fundamentals

Chapter 4.6: Evaluation with Confusion Matrix

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Outline

- 1 Introduction to Model Evaluation
- 2 The Confusion Matrix
- 3 Evaluation Metrics
- 4 Advanced Topics
- 5 Practical Considerations
- 6 Summary and Best Practices

Why Evaluate Machine Learning Models?

The Challenge

Building a model is not enough – we need to measure its performance!

Key Questions:

- How accurate is our model?
- What types of errors does it make?
- Is it better than alternatives?
- Can we trust it in production?

Evaluation Goals:

- Measure performance
- Compare models
- Identify weaknesses
- Guide improvements

Classification vs. Regression Evaluation

Problem Type	Output	Metrics
Classification	Discrete labels	Accuracy, Precision, Recall F1-score, Confusion Matrix
Regression	Continuous values	MSE, RMSE, MAE, R^2

Focus of This Lecture

We focus on **classification evaluation** using the confusion matrix and derived metrics.

What is a Confusion Matrix?

Definition

A **confusion matrix** is a table that visualizes the performance of a classification algorithm by comparing predicted labels against actual labels.

Purpose

- Shows where the model is "confused"
- Reveals types of errors made
- Foundation for computing evaluation metrics
- Applicable to binary and multi-class problems

Binary Classification Confusion Matrix

		Predicted	
		Positive	Negative
Actual		1	0
	Positive	1 TP	0 FN
	Negative	0 FP	1 TN

- **TP (True Positive):** Correctly predicted positive
- **TN (True Negative):** Correctly predicted negative
- **FP (False Positive):** Incorrectly predicted positive (Type I error)
- **FN (False Negative):** Incorrectly predicted negative (Type II error)

Example: Medical Diagnosis

Problem: Predicting if a patient has a disease

		Predicted	
		Disease	Healthy
Actual	Disease 100	1 85	0 15
	Healthy 100	0 8	1 92

Interpretation:

- TP = 85 (correct disease detection)
- FN = 15 (missed diseases!)
- FP = 8 (false alarms)
- TN = 92 (correct healthy)

Critical Question

Are 15 missed diseases acceptable?

Multi-class Confusion Matrix

For problems with $K > 2$ classes:

		Predicted			
		A	B	C	D
Actual	A	50	3	2	5
	B	4	45	1	4
	C	2	3	48	2
	D	3	2	3	52

- Diagonal elements: correct predictions
- Off-diagonal: misclassifications
- Each row sums to total actual instances of that class

Accuracy

Definition

Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Advantages:

- Simple, intuitive
- Easy to compute
- Good for balanced datasets

Limitations:

- Misleading with imbalanced data
- Doesn't show error types
- May hide poor performance

Medical Example

$$\text{Accuracy} = \frac{85+92}{200} = \frac{177}{200} = 88.5\%$$

The Imbalanced Data Problem

Scenario: Fraud detection (1% fraud rate)

		Predicted	
		Fraud	Normal
Actual	Fraud 10	0	10
	Normal 990	0	990

Naive Classifier

A model that always predicts "Normal" achieves:

$$\text{Accuracy} = \frac{0 + 990}{1000} = 99\%$$

But catches **zero frauds!**

Lesson

Accuracy alone is insufficient for imbalanced datasets!

Precision

Definition

Precision measures the accuracy of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Question: "Of all instances predicted as positive, how many are actually positive?"

High Precision means:

- Few false alarms
- High confidence in positive predictions
- Low Type I error

Important when:

- False positives are costly
- e.g., Spam detection
- e.g., Drug approval

Medical Example

$$\text{Precision} = \frac{85}{85+8} = \frac{85}{93} \approx 91.4\%$$

Recall (Sensitivity, True Positive Rate)

Definition

Recall measures the coverage of actual positives:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Question: "Of all actual positive instances, how many did we find?"

High Recall means:

- Few missed positives
- Good coverage
- Low Type II error

Important when:

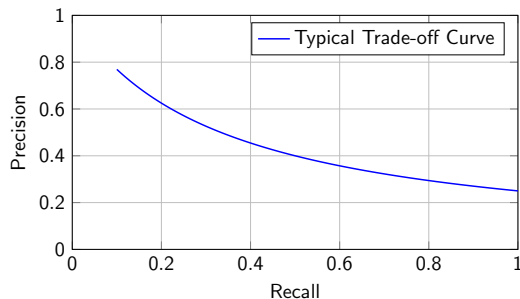
- Missing positives is costly
- e.g., Disease detection
- e.g., Security threats

Medical Example

$$\text{Recall} = \frac{85}{85+15} = \frac{85}{100} = 85\%$$

Missing 15% of diseases!

Precision vs. Recall Trade-off



The Trade-off

- Increasing recall often decreases precision
- Lowering threshold $\rightarrow$ more positives $\rightarrow$ higher recall, lower precision
- Raising threshold $\rightarrow$ fewer positives $\rightarrow$ higher precision, lower recall

F1-Score: Harmonic Mean

Definition

The **F1-score** is the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

Why harmonic mean?

- Penalizes extreme imbalance between precision and recall
- If either is very low, F1 is low
- Example: Precision=1.0, Recall=0.1 $\rightarrow$ F1=0.18 (not 0.55!)

Medical Example

$$\begin{aligned} F_1 &= 2 \cdot \frac{0.914 \times 0.85}{0.914 + 0.85} \\ &= 2 \cdot \frac{0.777}{1.764} \approx 0.881 = 88.1\% \end{aligned}$$

F_β -Score: Weighted Harmonic Mean

Definition

The F_β -score generalizes F1 with a weight β :

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Precision} \times \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

- $\beta = 1$: Equal weight (F1-score)
- $\beta < 1$: Favors precision (e.g., $F_{0.5}$)
- $\beta > 1$: Favors recall (e.g., F_2)

When to Use

- F_2 : Medical diagnosis (recall more important)
- $F_{0.5}$: Search ranking (precision more important)

Specificity and Other Metrics

Definition

Specificity (True Negative Rate):

$$\text{Specificity} = \frac{TN}{TN + FP}$$

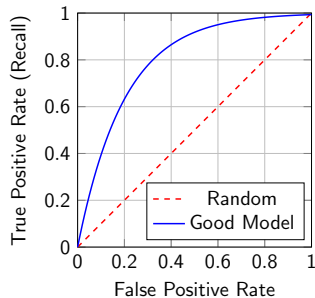
Other useful metrics:

- **False Positive Rate (FPR):** $\frac{FP}{FP+TN} = 1 - \text{Specificity}$
- **Negative Predictive Value (NPV):** $\frac{TN}{TN+FN}$
- **False Discovery Rate (FDR):** $\frac{FP}{FP+TP} = 1 - \text{Precision}$
- **Balanced Accuracy:** $\frac{\text{Sensitivity} + \text{Specificity}}{2}$

Medical Example

$$\text{Specificity} = \frac{92}{92+8} = 92\%$$

Receiver Operating Characteristic (ROC) Curve



Key Points:

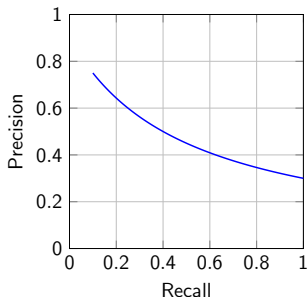
- Plots TPR vs FPR at various thresholds
- Diagonal = random classifier
- Upper-left = perfect classifier
- AUC = Area Under Curve

AUC Interpretation:

- 0.5: Random
- 0.7-0.8: Fair
- 0.8-0.9: Good
- 0.9-1.0: Excellent

Precision-Recall Curve

Alternative to ROC for Imbalanced Data



When to use:

- Highly imbalanced datasets
- Focus on positive class
- Care about precision

Advantages over ROC:

- Not affected by large number of true negatives
- More informative for rare events

Average Precision (AP)

Area under the Precision-Recall curve

Multi-class Metrics

Extending binary metrics to multi-class:

Macro-averaging

Compute metric for each class, then average:

$$\text{Macro-Precision} = \frac{1}{K} \sum_{i=1}^K \text{Precision}_i$$

Treats all classes equally (good for balanced classes)

Micro-averaging

Aggregate all TP, FP, FN across classes, then compute:

$$\text{Micro-Precision} = \frac{\sum_{i=1}^K TP_i}{\sum_{i=1}^K (TP_i + FP_i)}$$

Gives more weight to larger classes

Weighted-averaging

Weight each class by its support (number of instances)

Multi-class Example

Confusion matrix from earlier:

	A	B	C	D
A	50	3	2	5
B	4	45	1	4
C	2	3	48	2
D	3	2	3	52

Class A metrics:

- $TP_A = 50$
- $FP_A = 4 + 2 + 3 = 9$
- $FN_A = 3 + 2 + 5 = 10$
- $\text{Precision}_A = \frac{50}{59} = 84.7\%$
- $\text{Recall}_A = \frac{50}{60} = 83.3\%$

Macro-averaged F1:

- 1 Compute F1 for each class
- 2 Average: $F1_{\text{macro}} = \frac{F1_A + F1_B + F1_C + F1_D}{4}$

Cohen's Kappa

Definition

Cohen's Kappa (κ) measures agreement beyond chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o = observed accuracy, p_e = expected accuracy by chance

Interpretation:

- $\kappa < 0$: Agreement worse than chance
- $\kappa = 0$: Agreement equal to chance
- $0 < \kappa < 0.2$: Slight agreement
- $0.2 < \kappa < 0.4$: Fair agreement
- $0.4 < \kappa < 0.6$: Moderate agreement
- $0.6 < \kappa < 0.8$: Substantial agreement

$0.8 < \kappa < 1$: Almost perfect agreement

Use Case

Particularly useful for imbalanced datasets and inter-rater reliability

Choosing the Right Metric

Scenario	Recommended Metric
Balanced classes	Accuracy, F1-score
Imbalanced classes	Precision-Recall, F1, AUC-PR
False positives costly	Precision, Specificity
False negatives costly	Recall, Sensitivity
Need balanced view	F1-score, Matthews Correlation
Threshold-independent	ROC-AUC
Multi-class	Macro/Micro F1, Confusion Matrix

Domain-Specific Considerations

Medical Diagnosis:

- Prioritize Recall
- Missing disease = life-threatening
- Accept some false alarms
- Use F_2 score

Spam Detection:

- Prioritize Precision
- False positive = lost email
- Some spam acceptable
- Use $F_{0.5}$ score

Fraud Detection:

- High imbalance (rare frauds)
- Use Precision-Recall curve
- Cost-sensitive learning
- Consider business impact

Information Retrieval:

- Precision@K
- Mean Average Precision
- Normalized Discounted Cumulative Gain (NDCG) (ranking quality)

Cross-Validation and Confusion Matrix

Why Cross-Validation?

Single confusion matrix may not be representative

K-Fold Cross-Validation Approach:

- 1 Split data into K folds
- 2 For each fold:
 - Train on K-1 folds
 - Test on remaining fold
 - Compute confusion matrix
- 3 Aggregate K confusion matrices
- 4 Compute average metrics with confidence intervals

Reporting

Report: Mean $\pm$ Standard Deviation

Example: $F1 = 0.85 \pm 0.03$ (across 5 folds)

Common Pitfalls

1. Relying Solely on Accuracy

Always examine the confusion matrix and multiple metrics

2. Ignoring Class Imbalance

Use appropriate metrics and resampling techniques

3. Not Considering Costs

Different errors have different real-world costs

4. Testing on Training Data

Always use separate test set or cross-validation

5. Cherry-picking Metrics

Report all relevant metrics transparently

Key Takeaways

1 Confusion Matrix is Essential

- Foundation for understanding model performance
- Shows types of errors, not just overall accuracy

2 No Single Metric is Perfect

- Accuracy can be misleading
- Always consider precision, recall, and F1

3 Context Matters

- Choose metrics based on domain requirements
- Consider costs of different error types

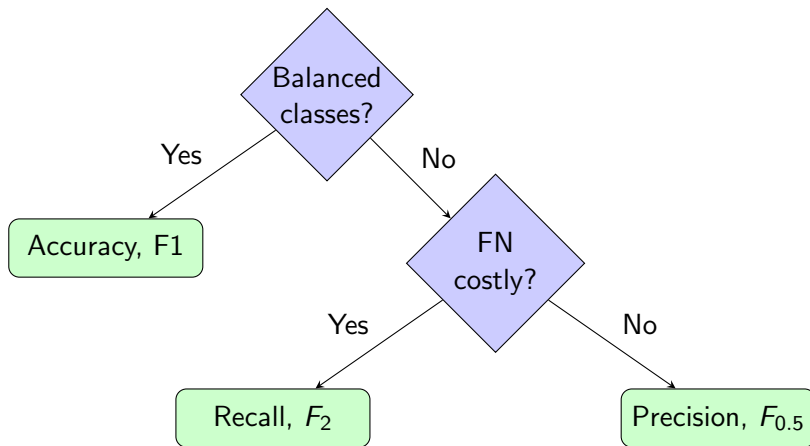
4 Handle Imbalanced Data Carefully

- Use appropriate metrics (PR-AUC, F1)
- Consider resampling or cost-sensitive learning

5 Report Multiple Metrics

- Transparency builds trust
- Allows for informed decision-making

Metric Selection Guide



Practice Problems

Problem 1: Given a confusion matrix, compute all metrics

120	30
20	180

Problem 2: A model achieves 99% accuracy on imbalanced data (1% positive class). Is it good? Why or why not?

Problem 3: You have two models:

- Model A: Precision=0.95, Recall=0.70
- Model B: Precision=0.75, Recall=0.95

Which would you choose for (a) spam detection, (b) cancer detection?

Problem 4: Implement confusion matrix computation from scratch

Thank You!

Questions?